

EMPLOYEE PERFORMANCE & WORKFORCE ANALYTICS

Year

All

Executive Workforce Overview

Total Employees
971

Total Salary
53.72M

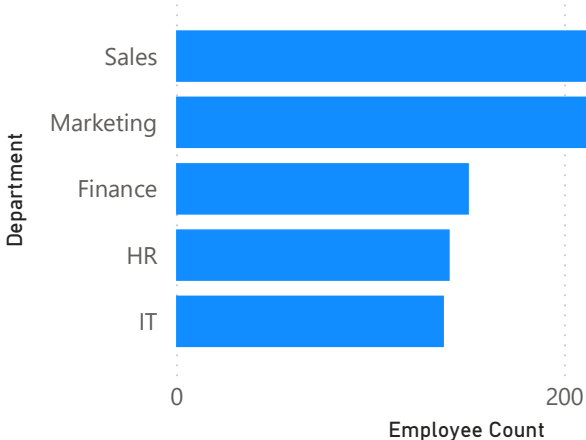
Average Salary
55.33K

Average Age
42.10

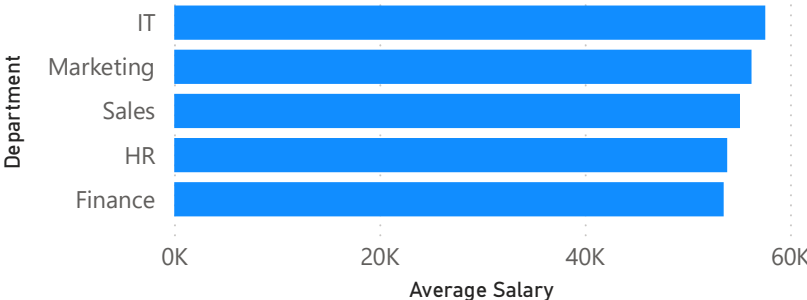
Average Performance
2.93

Highest Salary
112.79K

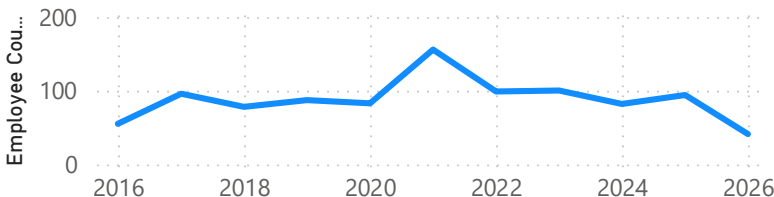
Employees by Department



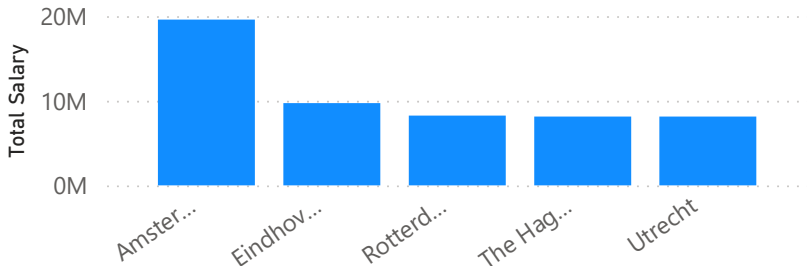
Average Salary by Department



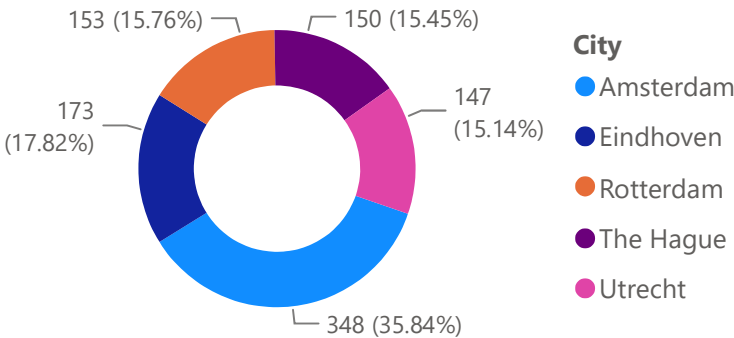
Employee Hiring Trend



Salary by City



Employees by City



Employee Performance Analysis

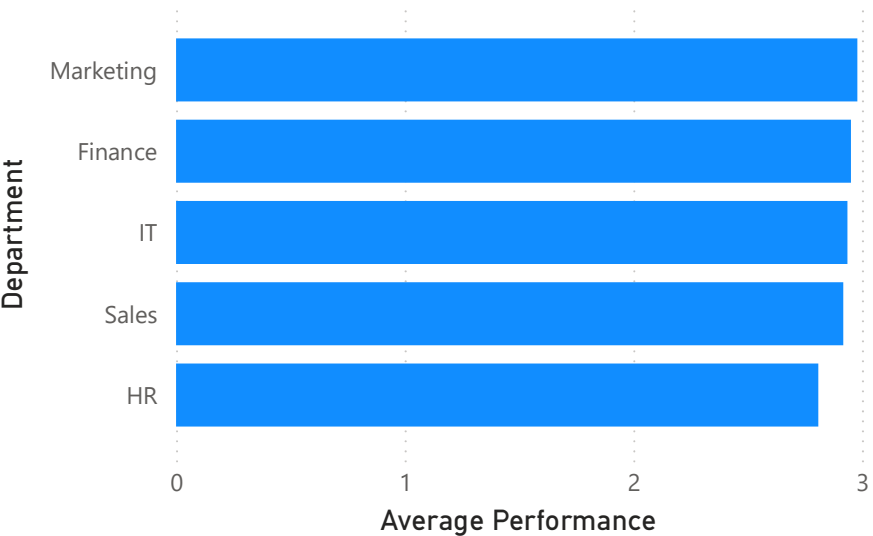
Total Employees
971

Average Performance
2.93

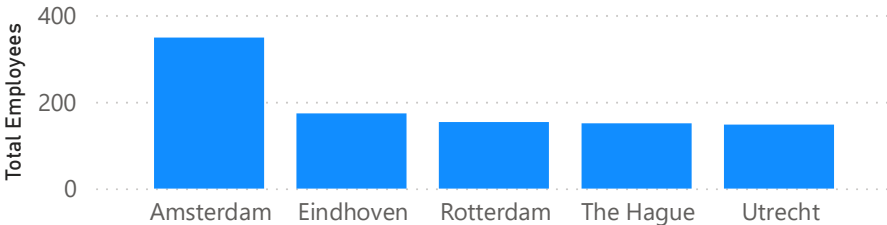
High Performers
215

Highest Salary
112.79K

Average Performance by Department

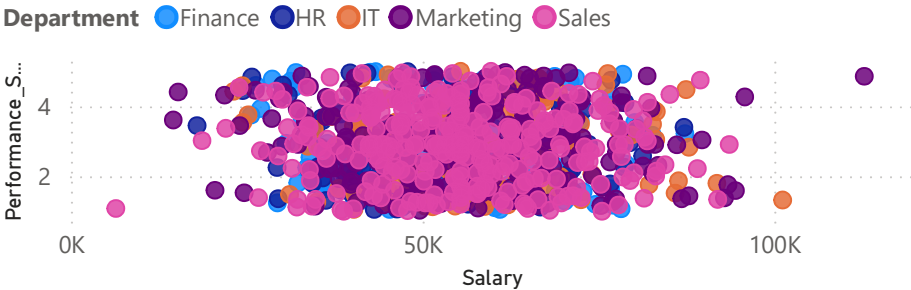


Employees by City



Employee_Id	Name	Departm
1	Dr. Todd Lester	Sales
10	Christopher Bailey	Marketir
100	Rachel Howell	Sales
1000	April Adams	HR
101	Jared Rivera	Sales
103	Crystal Walker	Finance
104	Melissa Gillespie	IT
Total		

First Employee_Id by Department, Salary and Performance_Score



WORKFORCE ANALYTICS USING PYTHON AND POWER BI

PROJECT OVERVIEW

- The project analyzes workforce data using Python and Microsoft Power BI to support business decision-making.
- A dataset containing **1,040 employee records** was assessed for quality issues before analysis.
- Python was used for data cleaning and preprocessing, while Power BI was used to build interactive dashboards.
- The analysis focused on workforce distribution, employee compensation, and performance across departments and locations.

DATA CLEANING PROCESS

- Explored the dataset using Python (*head()*, *info()*, *isnull().sum()*, and *duplicated().sum()*).
- Removed **40 duplicate records** to eliminate double counting.
- Removed **29 records with missing Employee IDs** because unique identifiers could not be recovered.
- Converted Age, Salary, and Performance Score to numeric data types.
- Replaced invalid ages, salaries, and performance scores using median imputation.
- Converted Join Date to datetime format.
- Standardized department and city names by removing extra spaces and applying consistent capitalization.
- Verified the cleaned dataset contained **971 valid employee records** with no duplicate or missing values.

DASHBOARD DEVELOPMENT

1. Imported the cleaned dataset into Microsoft Power BI.
2. Developed two interactive dashboards:
 - Executive Workforce Overview
 - Employee Performance Analysis
3. Included interactive slicers for Department and Year.
4. Created KPI cards for:
 - Total Employees
 - Total Salary
 - Average Salary
 - Average Age
 - Highest Salary
 - Average Performance Score

5. Developed charts showing:
 - Employee distribution by department
 - Average salary by department
 - Salary by city
 - Employee distribution by city
 - Salary versus performance
 - Departmental performance comparison.

BUSINESS ANALYSIS

Workforce Distribution

- The organization has 971 employees distributed across five departments and five cities.
- Sales is the largest department with 292 employees (30%).
- Marketing follows with 249 employees.
- Amsterdam is the largest workforce location, accounting for 35.8% of employees.

Compensation Analysis

- Total salary expenditure is approximately \$53.7 million.
- Average employee salary is \$55,327.
- IT has the highest average salary (\$57,589).
- Finance and HR have the lowest average salaries.
- Salary differences across departments are relatively small, indicating a fairly flat compensation structure.

Performance Analysis

- The average employee performance score is 2.93 out of 5.
- Marketing records the highest average performance score.
- HR records the lowest average performance score.
- The findings indicate opportunities for targeted performance improvement within HR.

Data Quality Assessment

- The raw dataset contained duplicate records, missing employee IDs, invalid ages, malformed salary entries, and inconsistent text formatting.
- These issues demonstrate the need for stronger data governance and validation controls before analysis.

RECOMMENDATIONS

- Review salary structures, particularly for IT and Finance, to maintain market competitiveness.
- Investigate HR's lower performance through targeted training and workload assessments.

- Evaluate staffing levels in Sales and Marketing to ensure workforce size aligns with organizational output.
- Implement mandatory validation rules for employee IDs, salaries, dates, and other critical fields.
- Standardize department and city naming conventions across all source systems.
- Conduct periodic data quality audits to improve reporting accuracy and reduce future data-cleaning efforts.

CONCLUSION

- Python successfully transformed a poor-quality workforce dataset into a reliable analytical dataset.
- Power BI converted the cleaned data into interactive dashboards that support workforce planning and strategic decision-making.
- The project identified key trends in workforce composition, compensation, and employee performance while emphasizing the importance of effective data governance for sustainable business intelligence.